REQUEST FOR ENGINEERING PROTOTYPE

Compact Integer Dot-Product Accelerator

Client brief | Rev 0.1 | 07 November 2026 | Engineering proof of concept

Engagement	Client-side details
Reference	OES-ACC-01
Requester	FPGA Platform Lead, Orion Edge Systems (fictional)
Target hardware	Digilent Nexys A7-100T, Artix-7
Budget / capacity	Hard cap: 50 engineering hours; no additional hardware purchase
Priority	Correctness and verification evidence first; performance second
Status	Exploratory brief — contractor to clarify before implementation

1 Business problem

Our edge sensor prototype computes small weighted sums in firmware. We want to evaluate a dedicated integer datapath as an offload candidate. This is NOT a request for a production neural-network accelerator or a CPU speedup claim. We need a reliable FPGA proof of concept and credible verification evidence before investing further.

The prototype must run on the Nexys A7-100T using onboard resources, with clear documentation for future reuse.

2 What we want it to do

For one job, the system receives four pairs of signed 8-bit integers (a0,b0) through (a3,b3) and produces the exact signed dot-product:

$$R = a0*b0 + a1*b1 + a2*b2 + a3*b3$$

The output shall represent the exact integer result without rounding or saturation. A 20-bit signed result is acceptable. Operands are two’s-complement values in the range -128 to +127.

3 Customer expectations (rough requirements)

ID	Expectation
R-01	Produce mathematically correct results for valid four-pair jobs, including negative operands and edge values.
R-02	Use a clocked, synthesizable SystemVerilog design. The client prefers a resource-conscious architecture over four unconditionally parallel multipliers.
R-03	Provide an explicitly documented input/output handshake. A stalled receiver must not cause a valid result to change or disappear.
R-04	Reset must return the design to a known idle state. Handling of an in-flight job at reset must be stated.
R-05	Implement on the Nexys A7-100T at 100 MHz and report actual post-implementation timing and resource usage.
R-06	Verify functionality with simulation and a self-checking reference model; demonstrate meaningful SVA/formal analysis.
R-07	Demonstrate an on-board PASS/FAIL test using no extra hardware.

4 Boundaries and engineering freedom

- You choose the microarchitecture, FSM, pipeline/register strategy, input framing and interface naming; explain your choices before RTL implementation.
- Aim for one outstanding job at a time. Multiple in-flight jobs, burst DMA, memory-mapped registers and software drivers are not required.
- No UART, AXI interconnect, DDR controller, high-level synthesis, floating point, configurable matrix dimensions, CRC, or external peripherals in this engagement.
- There is no minimum throughput requirement. Propose a realistic maximum latency in cycles, clearly stating when it starts and whether output backpressure is excluded.

5 Verification is a first-class deliverable

We are not satisfied by “simulation passes.” Please provide a concise verification plan, a reference model, directed corner cases, randomized scenarios where useful, and assertions that exercise the intended protocol and datapath behavior.

- At least three non-vacuous safety properties with reproducible formal results (e.g., reset/idle behavior, output stability during backpressure, no unauthorized result).
- Arithmetic integrity must be checked against an independent model. A reduced-width formal equivalence proof is welcome; full-width arithmetic formal proof is a stretch goal, not a condition of acceptance.
- State all environmental assumptions, proof bounds, tool limitations, unresolved failures, and uncovered behaviors. Do not label bounded evidence as an unconditional proof.

6 Acceptance demonstration

The handover will be accepted when the engineering lead can reproduce simulation, review verification evidence, generate a bitstream, and observe an on-board test that distinguishes pass from fail. The implementation report should show timing closure at the stated 100 MHz clock, or clearly identify a blocker and proposed remedy.

Client-provided directed samples are supplied separately as CSV; they are a minimum smoke test, NOT a complete test plan. Additional tests for reset, stalled interfaces, back-to-back jobs and signed arithmetic edge cases are expected.

7 Expected handover package

- RTL source, simulator testbench, independent reference model, assertions/formal setup, board top and Vivado constraints.
- Reproducible run instructions; one-page architecture description and state/interface diagrams.
- Verification summary: tests run, assertions attempted/passed, assumptions, coverage or non-vacuity evidence, limitations.
- FPGA evidence: build/tool version, target part, utilization, timing report, and brief demonstration notes or video.
- A short issue log and the next three technical risks or improvements, prioritized by engineering value.

8 Engagement rules and checkpoints

Checkpoint	What the contractor submits	Client review
M0 - before coding	Assumptions, key questions, interface sketch, feasibility and hour estimate	Approve scope and unresolved choices
M1 - functional RTL	Buildable RTL and self-checking simulation results	Review behavior and corner cases
M2 - verification	SVA/formal artifacts, evidence, known limits	Review confidence, not just PASS labels
M3 - final hardware	Board demo and full source/report package	Reproduce and accept or request defect fixes

Hour planning guidance: clarification/spec 7 h; RTL 10 h; simulation/reference 9 h; formal 10 h; FPGA demonstration 7 h; handover 3 h; contingency 4 h. Total cap: 50 h. If any requirement threatens the cap, notify the client and propose scope cuts before proceeding.

9 Decisions we expect you to raise

This brief is intentionally incomplete. Before implementation, send us a one-page technical response identifying assumptions, risks and questions. In particular, think about the input transaction boundary, handshake semantics, reset during a job, signed arithmetic sizing, result holding, and how to demonstrate correct operation on the board. We expect engineering judgment, not just literal implementation.

10 Reference material provided

Digilent Nexys A7-100T reference manual: https://digilent.com/reference/_media/reference/programmable-logic/nexys-a7/nexys-a7_rm.pdf

Digilent board master XDC: https://github.com/Digilent/digilent-xdc/blob/master/Nexys-A7-100T-Master.xdc

AMD Vivado UG901 (DSP and MAC inference): https://docs.amd.com/r/en-US/ug901-vivado-synthesis/Multiply-Add-and-Multiply-Accumulate

Training note: This is a fictional client brief created for a self-directed engineering exercise. It is not an actual contract or evidence of paid client work.